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B. E. (Seventh Semester) Examination, Nov.-Dec., 2015

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(Old Scheme)

(CSE, IT Engg. Branch)

CRYPTOGRAPHY & NETWORK SECURITY

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

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*Note : Part (a) of each question is compulsory and contains 2 marks. Attempt any **two** part from (b), (c) and (d) each part carry 7 marks.*

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1. (a) Find gcd (56, 86) using Euclidean algorithm.

(b) Describe the Play Fair cipher matrix making and encryption rules, using "Encryption Algorithm" as the key and encrypting the message "Restricted Space Environments".

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(c) Draw the extended Euclid's process model and also write the algorithm with an appropriate example.

(d) Describe DES (Data Encryption Standards) scheme and also the working of each block associated with it.

2. (a) What are the two basic functions used in encryption algorithms?

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(b) How AES is used for encryption/decryption? Discuss with example.

(c) Explain the RC5 stream cipher method used for encryption and decryption.

(d) Draw the idea ("International Data Encryption Algorithm") rounds structures and explain in detail.

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3. (a) Usine Fermats theorem, find 3^{201} mod 11.

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(b) User A and B use the Diffie Hellman key exchange technique a common prime $p = 11$ and a primitive root $g = 7$.

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(i) If user A has private key Pra = 3 what is A's public key (Pua)?

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(ii) If user B has private key Prb = 6 what is B's public key (Pub)?

(iii) What is the shared secret kety (Ks)?

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(iv) How man in middle attack can be performed in Diffie Hellman algorithm.

(c) How does SHA-1 logic produce message digest?

(d) Sketch the X.509 formats and explain the fields. Explain about authentication procedures supported in X.509.

4. (a) Find the solution of congruence $3x \equiv 4 \bmod 8$.

(b) Briefly explain the Kerberos version 5 messages exchanges and give the over view of Kerberos using a neat sketch.

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(c) What are the two types of digital signature? Explain each briefly. Also explain the digital signature algorithm and the digital signature system.

(d) How does PGP provide confidentiality and authentication service for e-mail and file storage applications? Draw the block diagram and explain its components.

5. (a) Define the term Hardened firewall.

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(b) Write the classification of virus. Explain in brief any three.

(c) What is fire-wall? Explain in detail about firewalls, characterisitcs and types with needed block diagrams.

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(d) Define intrusion detection and explain the different types of detection mechanisms in detail.

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